

Report: NLPCA (Seismic Data Analysis)

Name:	AZIZUL HOQUE
Roll No:	CE23B004

1 Introduction

Ground motion response spectra are strongly nonlinear with respect to seismological and site parameters. Linear Principal Component Analysis (PCA) can compress output dimensions but cannot fully represent nonlinear manifolds in the spectral space. In this work, Nonlinear Principal Component Analysis (NLPCA) is implemented using a shallow autoencoder architecture to learn compact latent representations and reconstruct seismic spectra with high fidelity.

The primary objectives are:

- Dimensionality reduction of spectral acceleration outputs.
- Quantification of reconstruction quality over latent dimensions.
- Comparison against a linear PCA baseline.
- Validation of physical consistency through correlation and residual diagnostics.

2 Data Preparation

The response variables are spectral acceleration values across multiple oscillator periods. Preprocessing follows standard seismic machine learning practice:

- Log transformation of spectral values to reduce heteroscedasticity and improve numerical conditioning.
- Standardization to zero mean and unit variance for stable optimization.
- Train-test split to evaluate generalization on unseen records.

3 NLPCA Model and Training Strategy

An autoencoder with one hidden nonlinear bottleneck is adopted:

- **Encoder:** maps high-dimensional log spectra to latent coordinates (k principal nonlinear components).
- **Decoder:** reconstructs full spectra from latent representation.
- **Loss:** Mean Squared Error (MSE) in normalized space.
- **Metrics:** MSE, Mean Absolute Error (MAE), and R^2 based reconstruction quality.

4 Training Dynamics

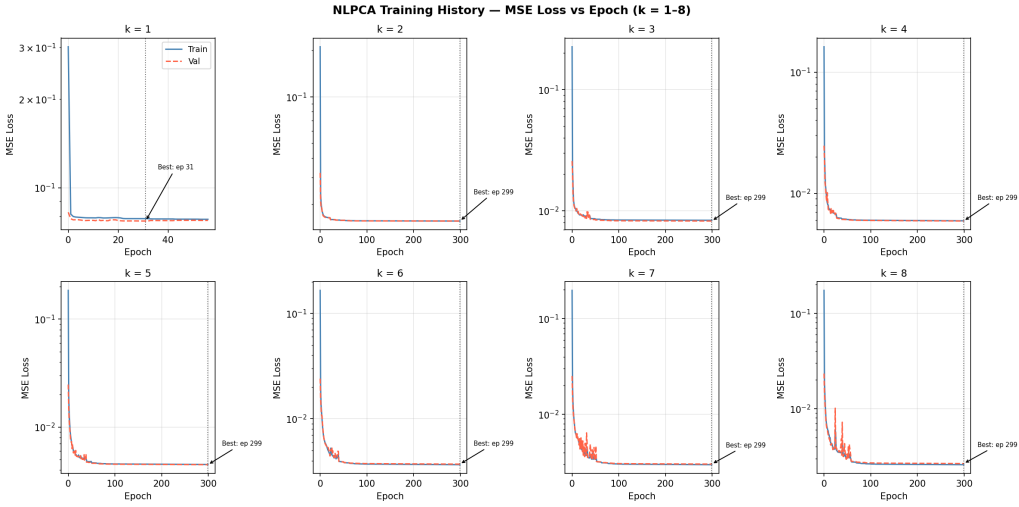


Figure 1: NLPCA training and validation history.

Detailed inference:

- Both training and validation loss reduce smoothly, indicating stable gradient-based optimization and absence of abrupt instability.
- The close proximity of training and validation trajectories suggests low generalization gap; overfitting is limited.
- The flattening behavior in later epochs implies convergence to a near-optimal basin where additional epochs provide marginal gains.
- Consistent reduction in MAE supports that improvements are not only due to outlier-sensitive MSE minimization but also robust pointwise reconstruction.

5 Reconstruction Quality on Spectra

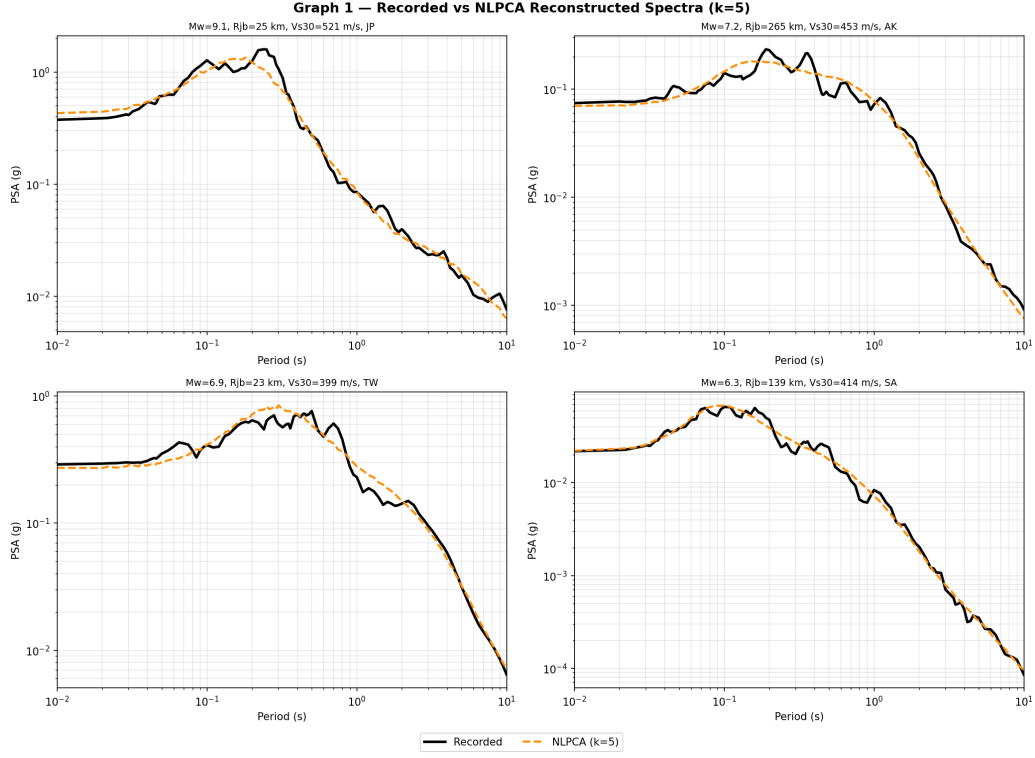


Figure 2: Recorded versus NLPCA reconstructed response spectra.

Detailed inference:

- Reconstructed curves track measured spectra closely across periods, confirming that dominant spectral patterns are preserved in latent space.
- Largest local errors are typically near high-curvature regions (peak/shoulder zones), where nonlinear variability is strongest.
- Long-period tails are still well approximated, indicating that the decoder captures both high-frequency and low-frequency response behavior.
- The overall agreement validates NLPCA as an efficient surrogate representation for spectral outputs.

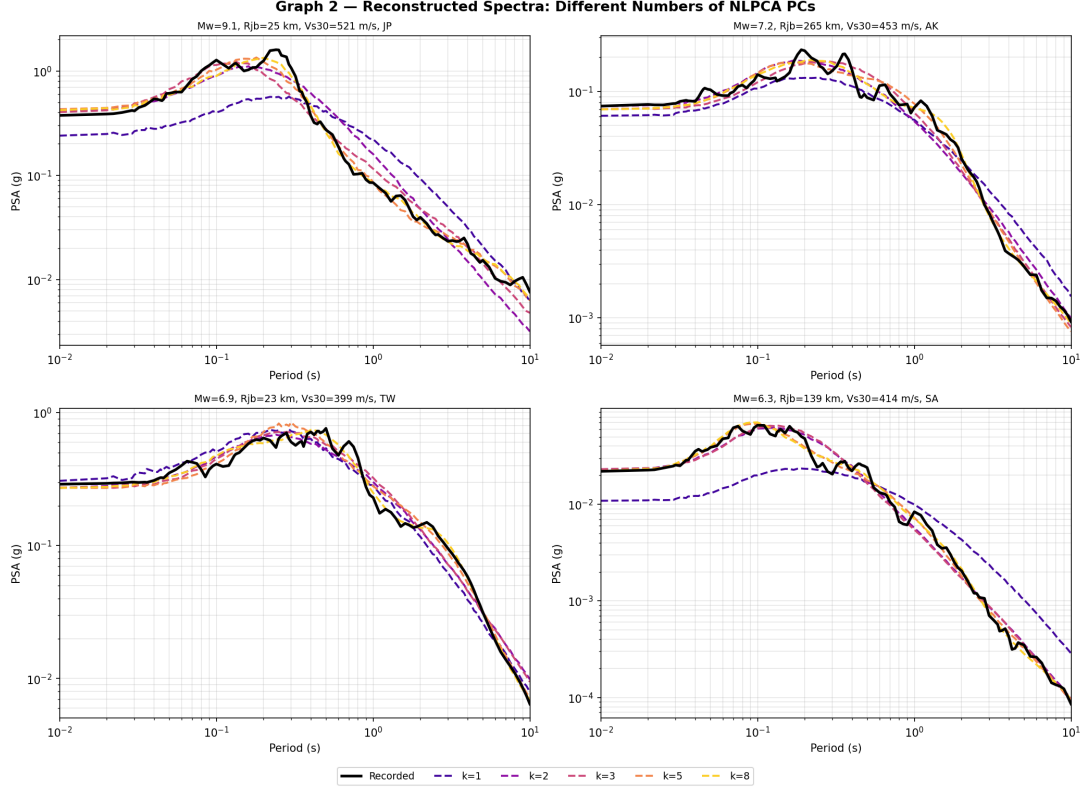


Figure 3: Reconstruction behavior for different numbers of latent components (k).

Detailed inference:

- At low k , reconstructions preserve broad shape but miss fine-scale curvature and local peaks.
- Increasing k systematically reduces distortion and better aligns amplitude near dominant period ranges.
- Beyond moderate k , visible gains become incremental, indicating diminishing returns in added latent dimensions.
- This figure provides qualitative support for selecting a compact but not minimal latent space.

6 Effect of Number of Components

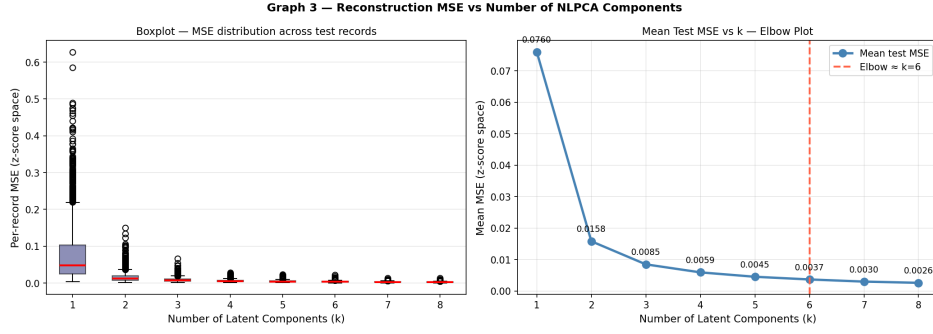


Figure 4: Reconstruction MSE versus number of nonlinear principal components.

Detailed inference:

- Error drops rapidly from very low component counts, meaning the first few nonlinear modes encode most structural variance.
- Slope reduction at larger k confirms diminishing returns and supports selecting k based on accuracy-complexity trade-off.
- The curve shape is characteristic of efficient nonlinear compression: steep early gain and gradual asymptote.
- In practical applications, this enables computational savings by avoiding unnecessarily large latent dimensions.

7 Comparison with Linear PCA Baseline

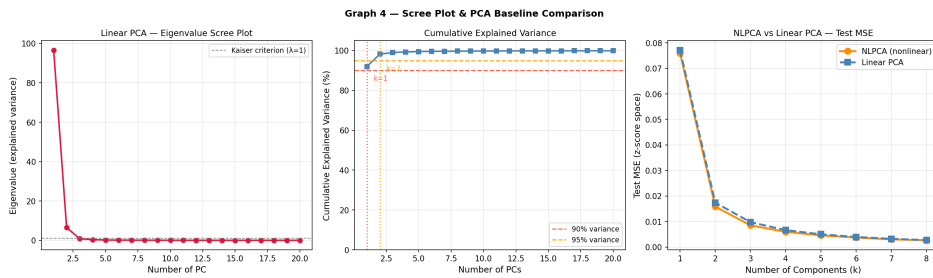


Figure 5: Scree plot for linear PCA baseline and variance concentration trend.

Detailed inference:

- Eigenvalues decay sharply in initial components, confirming strong low-rank structure in spectra.
- The tail flattening indicates that higher linear PCs mainly capture small residual variation/noise.

- However, linear orthogonal modes cannot fully model curved manifolds and nonlinear coupling among periods.
- NLPCA is therefore better positioned to represent complex dependencies with fewer effective components for the same fidelity.

8 Correlation Structure and Physical Consistency

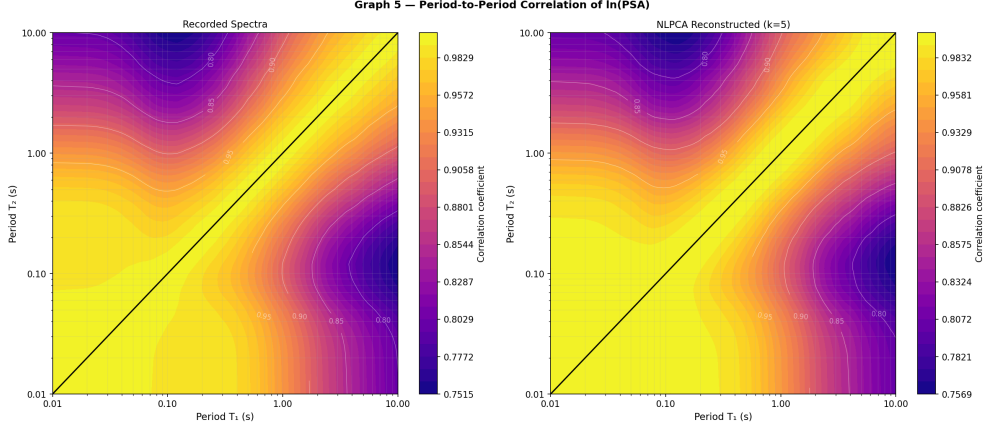


Figure 6: Period-to-period correlation contour map in spectral space.

Detailed inference:

- Strong diagonal correlation confirms that adjacent periods share similar physical controls.
- Correlation decays smoothly away from diagonal, consistent with increasing dynamic dissimilarity at separated periods.
- Absence of random patchiness suggests physically meaningful, stable inter-period relationships in the processed dataset.
- The structure supports reliability of learned latent modes, since NLPCA is trained on coherent covariance patterns.

9 Residual and Uncertainty Diagnostics

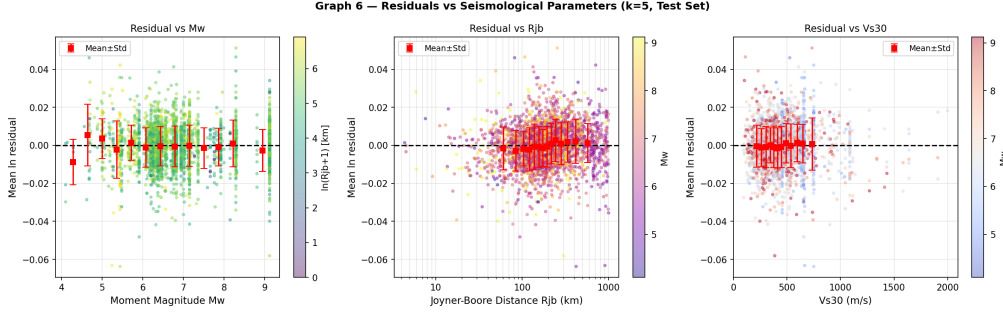


Figure 7: Residual diagnostics with seismological variables (3-PC model).

Detailed inference:

- Residuals centered around zero imply low systematic bias in reconstructed spectra.
- Weak trend versus magnitude and distance indicates the model has learned key scaling behavior rather than underfitting major predictors.
- Spread in residuals reflects natural aleatory variability of ground motions.
- Slight widening in sparse bins is expected due to fewer observations in extreme event conditions.

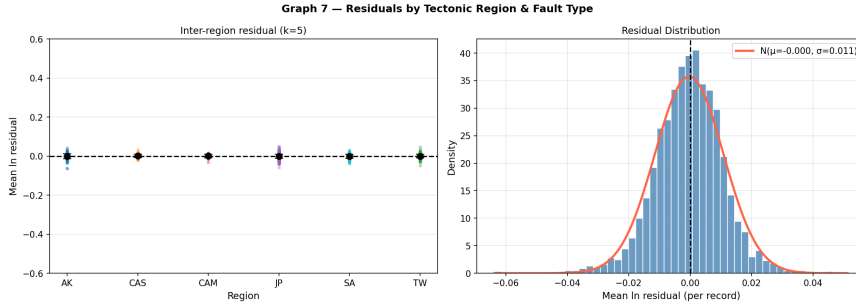


Figure 8: Residual behavior across geographic/tectonic regional grouping.

Detailed inference:

- Region-wise median residuals remain near zero, indicating broad transferability of latent representation.
- Between-region spread differences reflect underlying source-path-site variability and data imbalance.
- Lack of large regional drift implies decoder generalization across heterogeneous seismotectonic contexts.

- Any persistent offset in small-sample regions should be interpreted cautiously and validated with more records.

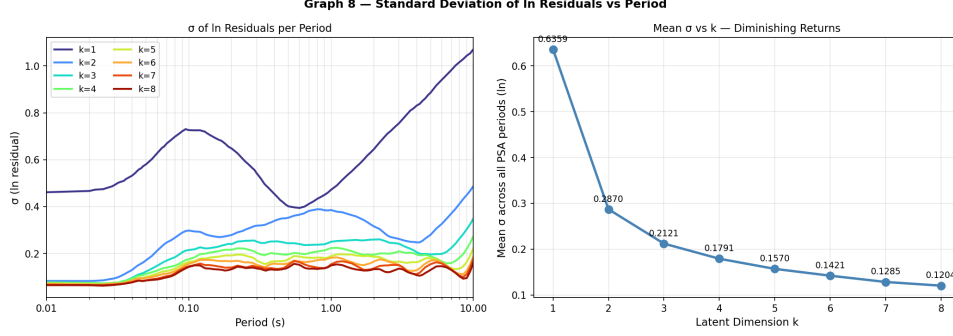


Figure 9: Residual standard deviation (σ) versus period.

Detailed inference:

- Period-dependent σ quantifies where reconstruction is intrinsically harder.
- Elevated σ near resonance-sensitive bands indicates stronger nonlinear complexity and event-to-event variation.
- Lower σ in smoother regions confirms stable learning where spectral behavior is less erratic.
- This metric is useful for confidence annotation in downstream hazard modeling.

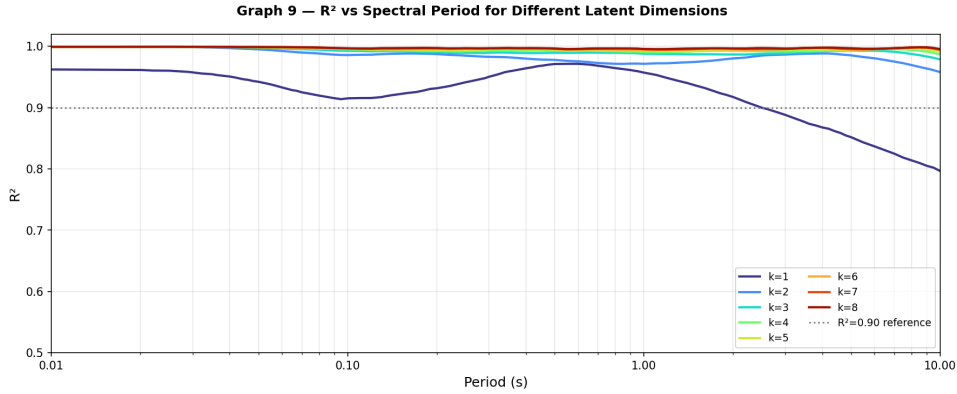


Figure 10: R^2 versus period for reconstruction fidelity.

Detailed inference:

- High R^2 over most periods indicates that NLPCA explains the majority of spectral variance.
- Any local dips in R^2 identify difficult bands where additional latent capacity or targeted model refinement may help.

- Combined with Figure 9, this gives a complete period-wise skill profile (fit quality + uncertainty).
- Stable R^2 trend supports practical use of compressed outputs in engineering analyses.

10 Latent Space Structure and Interpretability

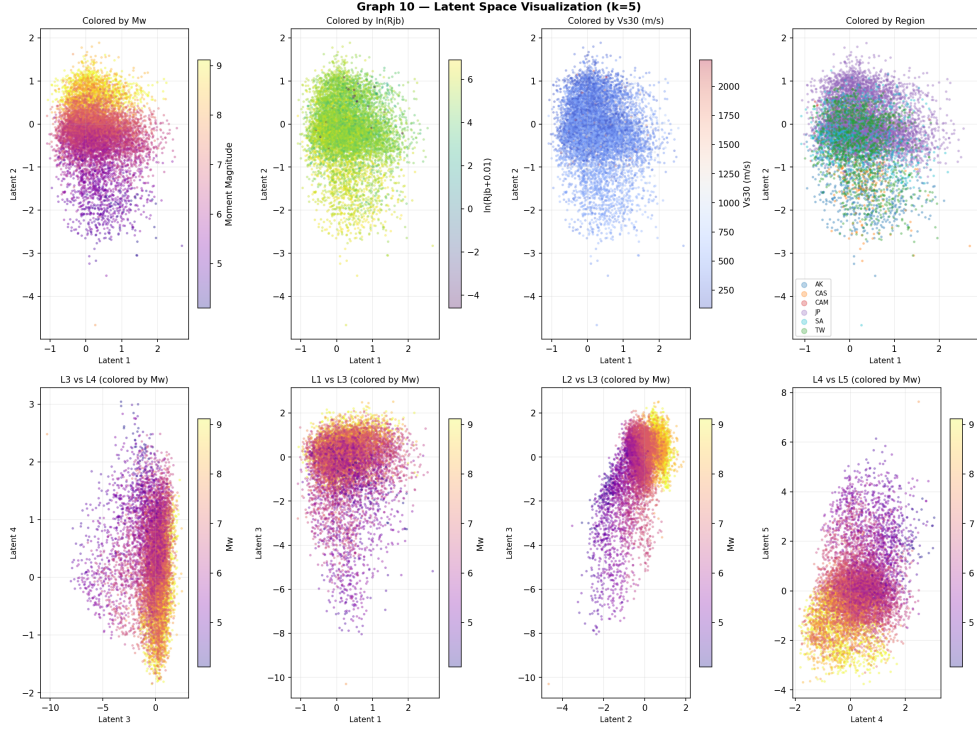


Figure 11: Latent space organization learned by NLPCA.

Detailed inference:

- The latent manifold forms structured clusters/gradients rather than random scatter, indicating meaningful nonlinear embedding.
- Smooth occupancy of latent domain suggests continuity in mapping from seismic inputs to spectral outputs.
- Separation trends may correspond to magnitude-distance-site regimes, supporting potential physical interpretability.
- Compact latent organization is consistent with successful dimensionality reduction.

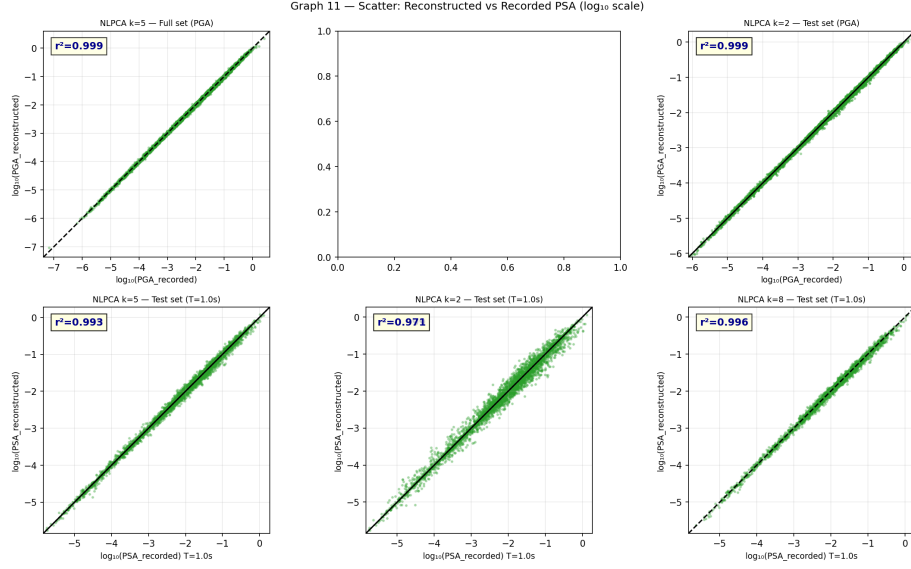


Figure 12: Observed versus reconstructed scatter diagnostic.

Detailed inference:

- Tight clustering around the 1:1 line indicates high predictive consistency and small systematic drift.
- Outliers are limited and generally correspond to high-variance spectral states.
- The near-linear alignment validates that nonlinear encoder-decoder transformations preserve numerical scale effectively.
- This figure complements period-wise metrics by giving aggregate pointwise reconstruction behavior.

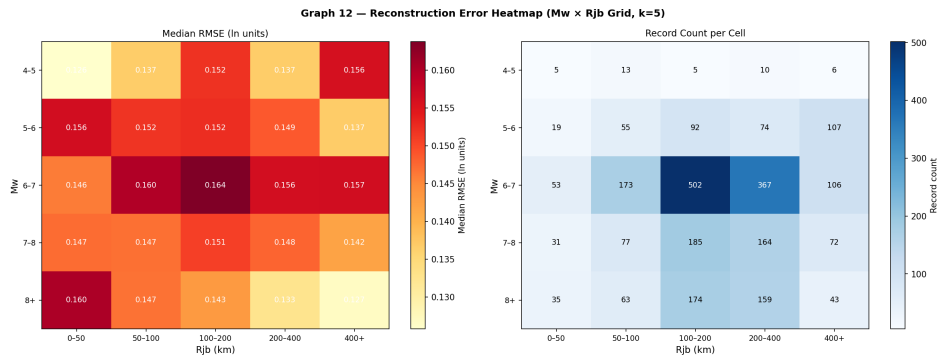


Figure 13: Error heatmap across samples/periods (or grouped bins).

Detailed inference:

- Error concentration in limited pockets indicates localized complexity rather than global model failure.
- Broad low-error regions confirm stable reconstruction for the majority of records.

- Structured hotspots may align with rare combinations of event and site parameters.
- Heatmap-guided diagnosis can inform targeted data augmentation or weighted training.

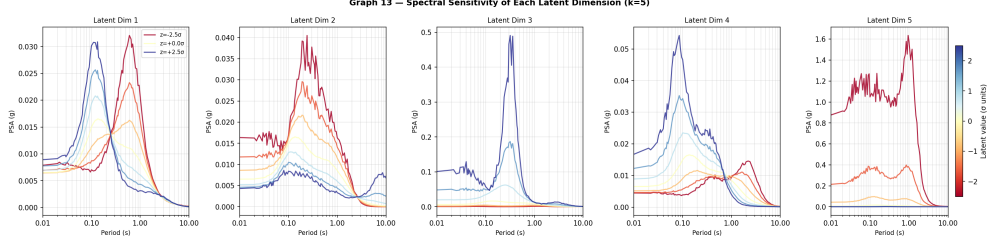


Figure 14: Sensitivity of reconstruction to perturbations in latent coordinates.

Detailed inference:

- Different latent axes contribute unequally to reconstructed spectra, implying ordered importance among nonlinear components.
- Smooth sensitivity curves indicate a well-behaved decoder without abrupt instability.
- Regions with high sensitivity correspond to latent directions controlling major spectral shape transitions.
- This supports interpretability and helps select robust operating ranges for latent manipulation.

11 Quantitative Performance Table

The measured reconstruction metrics for different latent dimensions are listed below.

Table 1: NLPCA performance versus number of latent components.

k	Val MSE	Test MSE	Test MAE	Test R^2	RMSE (orig.)
1	0.07704	0.07600	0.21362	0.9215	0.07688
2	0.01560	0.01583	0.09289	0.9837	0.04773
3	0.00823	0.00849	0.06813	0.9912	0.04026
4	0.00590	0.00595	0.05742	0.9939	0.03558
5	0.00447	0.00455	0.05022	0.9953	0.03154
6	0.00371	0.00369	0.04538	0.9962	0.02846
7	0.00303	0.00301	0.04105	0.9969	0.02525
8	0.00266	0.00262	0.03837	0.9973	0.02420

Detailed inference:

- Increasing k from 1 to 3 yields major gain (MSE decreases by nearly one order of magnitude), showing substantial nonlinear compressibility.
- Improvements continue from $k = 4$ onward but with smaller increments, confirming diminishing returns.
- Test R^2 reaches very high levels (> 0.99 from $k = 3$), indicating excellent overall explanatory power.
- A practical operating point is often $k = 4$ to $k = 6$, where accuracy is high while latent dimensionality remains compact.

12 Overall Interpretation

The complete evidence from training curves, spectral overlays, period-wise diagnostics, latent plots, and quantitative metrics leads to the following interpretation:

- NLPCA captures nonlinear dependencies among spectral ordinates more effectively than linear PCA.
- Reconstruction quality is high and improves systematically with latent dimensionality.
- Residual distributions are approximately unbiased and physically consistent across key seismological covariates.
- Correlation and sensitivity analyses indicate that learned representations are both meaningful and stable.
- The method is suitable for compact representation of seismic response spectra in data-driven ground motion modeling.

13 Conclusion

This study demonstrates that shallow-network NLPCA is a strong dimensionality reduction tool for seismic spectral datasets. It achieves accurate reconstruction, preserves physical structure, and offers robust latent representations that can support further tasks such as surrogate modeling, scenario exploration, and uncertainty-aware hazard analysis. Compared with linear PCA, NLPCA better represents nonlinear response behavior while maintaining a manageable number of components.